

CUSTOMER

PROJECT		x	
MANUFACTURER			
DOCUMENT NO.		x	
DOC TITLE		Ampacity Calculation	
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Unit	Reference	Clause	Symbol
			Formula
			TREFOIL GNDTREFOIL DUCTTREFOIL AIR

DESIGN DATA

Bonding Type	Design data		solid		solid	solid
Cond Mat	Design data		Al		Al	Al
Cond size	mm2	Design data	185		185	185
Cond design	Design data		CC		CC	CC
Cond Dn	mm	Design data	Dc	15.9		15.9
Cond Sc Dn	mm	Design data	Ds	17.6		17.6
Ins Mat	Design data		XLPE		XLPE	XLPE
Ins Dn	mm	Design data	Di	33.9		33.9
Ins Sc Dn	mm	Design data	dx	37.6		37.6
Ins Tn	mm	Design data	Ti	9.19		9.19
Core Dn	mm	Design data	Do	--		--
Lay-up OD	mm	Design data	Da	--		--
Cable Dn	mm	Design data	De	43.4		43.4
Core No	nos	Design data	n	1	1	1
Wire No	nos	Design data	nw	37		37
Cable weight	kg/km	Design data	M	6731		6731
Max cond temp	degC	Design Data	Omax	90		90
Earth fault current	A	Design Data	Ix/1sec	1	1493	1493
Rated voltage	kV	IEC60502-2	Table 1	U	33	33
Earth voltage	kV	IEC60502-2	Table 1	U0	19.053	19.053
Impulse voltage	kV	IEC60502-2	Table 14	Up	170	170

PARAMETER SETTING

Soil TR	K.m/W	Input	p	1.2	1.2	1.2	1.2
Multicore factor		Input	fm	1	1	1	1
Frequency	Hz	Input	f	50	50	50	50
Ambient temp.	degC	Input	Oa	30	30	40	40
Depth	mm	Input	L	1200	1000	1000	1000
Duct Type		Input			HDPE	HDPE	HDPE
External diameter of duct	mm	Input	dc	160	110.0	110.0	110.0
Thickness of duct	mm	Input	Td	0	0	5.0	5.0
Duct temperature	degC	Input	Om	85	85	85	85
Core space factor		Input	fs		1	1	1
Axial Circ distance H	mm	Input	Sh	500	0	0	0
Axial Circ distance V	mm	Input	Sv	500	0	0	0
Horizontal circuit	nos	Input	nh	1	1	1	1
Vertical circuit	nos	Input	nv	1	1	1	1
Temp. rise, ambient to max.	degC	IEC60287-1-1	1.4.1.1	dO	Omax - Oa	60	60
Core temperature, Oc	degC	Iteration		Oc		80	78
Axial Core distance	mm			S		44	160
Cable space	mm			S'		0	116
Circuit spacing H	mm			S'h		0	0
Circuit spacing V	mm			S'v		0	0
Axial distance cond to cab	mm			c		0	0
Ins Tn between cond	mm			t		0	0
Circumscribing radius cond	mm			r1		0.00	0.00
Coefficient belted cable		IEC60287-2	2.1.1.2	F2	1+3t/[2p(dx+t)-t]	0.00	0.00
Geometric factor		IEC60287-2	2.1.1.2	G	3F2ln(da/2r1)	0.000	0.000
Wire factor		BICC	Table 2.1	fw		0.0528	0.0528
Temp coeff - conductor		IEC60287-1-1	Table 1	a20		0.004	0.004
Ins Relative permittivity		IEC60287-1-1	Table 3	er		2.5	2.5
Loss factor		IEC60287-1-1	Table 3	tand		0.004	0.004
Dielectric loss	W/m/ph	EC60287-1-1	2.2	Wd	wCU02tand	0.000	0.000

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TREFOIL GNDTREFOIL DUCTTREFOIL AIR

DC RESISTANCE								
DC resistant at 20degC	ohm/m	IEC60228 Class 2		R0		1.640E-04	1.640E-04	1.640E-04
DC resistant at ambient	ohm/m	IEC60287-1-1	2.1.1	R0amb	$R0(1+a20(Ox-20))$	1.706E-04	1.706E-04	1.706E-04
DC resistant at 90degC	ohm/m	IEC60287-1-1	2.1.1	R0max	$R0(1+a20(Omax-20))$	2.103E-04	2.103E-04	2.103E-04

AC RESISTANCE								
Strand constant		IEC60287-1-1 Table 2		ks		1.0	1.0	1.0
Strand constant		IEC60287-1-1 Table 2		kp		0.8	0.8	0.8
Spacing correction		Mc Graw Hill Page 161				1.250	1.250	1.250
Bessel function		IEC60287-1-1	2.1.2	Xs4	$[ks(8pf/R0)10-7]2$			
Skin effect		IEC60287-1-1	2.1.2	Ys	$Xs4/(192+0.8Xs4)$			
Proximity function		IEC60287-1-1	2.1.4	Xp4	$[kp(8pf/R0)10-7]2$			
Proximity function		IEC60287-1-1	2.1.4	Yp"	$Xp4/(192+0.8Xp4)$			
Proximity effect		IEC60287-1-1	2.1.4	Yp	$Yp"(Dc/s)2[0.312(Dc/s)2+1.18/(Yp"+0.27)]$			
AC resistant at 20degC	ohm/m	IEC60287-1-1	2.1	R20	$R0(1+Ys+Yp)$			
AC resistant at 20degC - isolated	ohm/m	IEC60287-1-1	2.1	R'20	$R0(1+Ys)$			
AC resistant at 90degC	ohm/m	IEC60287-1-1	2.1	R90	$R0max(1+Ys+Yp)$	2.108E-04	2.107E-04	2.108E-04
AC resistant at 90degC - isolated	ohm/m	IEC60287-1-1	2.1	R'90	$R0max(1+Ys)$			

SHEATH LOSS FACTOR								
Equivalent mean diameter	mm	Design data		ds				
Equivalent resistant at 20degC	ohm/m	IEC60287-1-1		R20	p/A	3.448E-04	3.448E-04	3.448E-04
Equivalent resistant at ambient	ohm/m	IEC60287-1-1		Rx	$R20(1+a20(Ox-20))$			
Equivalent resistant at Oc	ohm/m	IEC60287-1-1		Rs	$R20(1+a20(Oc-20))$			
Sheath Inductance	H/m	IEC60287-1-1	2.3.3	L	$2ln(2s/Dm)x10-7$			
Sheath Reactance	ohm/m	IEC60287-1-1	2.3.3	X	$2wln(2s/Dm)x10-7$			
Circulating current loss factor		IEC60287-1-1	2.3.1	lam1'	$Rs/R/(1+(Rs/X)2)$			
Eddy current loss factor		IEC60287-1-1	2.3.6.1	lam1"		0.000E+00	0.000E+00	0.000E+00
Sheath loss factor		IEC60287-1-1	2.3	lam1	$lam1'+lam1"$	0.033	0.134	0.033

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THERMAL RESISTANCE

Thermal Resistant Insulation	K.m/W	IEC60287-2-1		T1	$(p/2p)LN[1+(2t/dc)]$	0.451	0.451	0.451
Thermal Resistant filler	K.m/W	IEC60287-2-1		T2f		0.000E+00	0.000E+00	0.000E+00
Thermal Resistant Bedding	K.m/W	IEC60287-2-1		T2		0.000E+00	0.000E+00	0.000E+00
Thermal Resistant Outercovering	K.m/W	IEC60287-2-1		T3	$(p/2p)LN[1+(2t/D)]$	0.208	0.130	0.130
Thermal Resistant External	K.m/W	IEC60287-2-1	2.2	T4		2.231	2.377	1.011

CURRENT RATING

Critical Temp Rise of Soil	degC	IEC60287-1-1		dO	$Wk(pc/2p)LN(Gf)$	0.000E+00	0.000E+00	0.000E+00
Top Equation		IEC60287-1-1	1.4.1	Etop	$dO-Wd(0.5T1+n(T2+T3)+T4)-dO$	59.9	59.9	50.0
Bottom Equation		IEC60287-1-1	1.4.1	Ebot	$RT1+nR(1+l\text{am}1)T2+nR(1+l\text{am}1+l\text{am}2)(T3+T4)$	3.242E-04	6.939E-04	3.438E-04
Calculated Amps	A	IEC60287-1-1	1.4.1	I	$\text{sqrt}(Etop/Ebot)$	309	294	381
Calculated MVA	MVA	IEC60287-1-1	1.4.1	MVA	$\text{sqrt}3*U*I$	18	17	22

ELECTRICAL DATA

Inductance	uH/m	IEC60287-1-1	2.3	L	$0.2LN(2s/Dm)$			
Reactance	uohm/m	IEC60287-1-1	2.3	X	$w*Lm*10^{-3}$			
Pos Seq Resistance	uohm/m			R1	$R'1(1+l\text{am}1+l\text{am}2)$			
Pos Seq Reactance	uohm/m			X1	$w*L*10^{-3}$			
Pos Seq Impedance	uohm/m			Z1	$\text{sqrt}(R12+Xm2)$			
Zero Seq Resistance	uohm/m			R0	$Rx+(n*R'x)$			
Capacitance	uF/km	IEC60287-1-1	2.2	C	$e/(18LN(Di/ds))$	2.119E-10	2.119E-10	2.119E-10
Charging current - Max	A/km	BICC Page 170		Ic max	$1.08*w*C*U0$			

THERMAL LOSS

Thermal Resistant Total	K.m/W	Mc Graw hill	Page 34	SumT	$T1+T2+T3+T4$			
Thermal loss	W/m	Mc Graw hill	Page 34	W	$(Oc-Oa)/\text{SumT}$			
Losses - Conductor	W/m	IEC60287-1-1	2.4.2	Wc	$3(I2R)$	20.18	18.20	30.64
Losses - Dielectric	W/m	IEC60287-1-1	2.2	Wd	$3*w*C*U02*\text{tand}$	0.000E+00	0.000E+00	0.000E+00
Losses - Metal sheath	W/m		IEC60287-1-1	Wm	$Wc*l\text{am}1$			

INDUCED PARAMETER

Normal - Sheath voltage	V/m	IEEE Std 575	Page 29	Us	$I*X$			
Normal - Circulating current	A	IEEE Std 575	Page 29	Is	$\text{sqrt}(I2(R/Rs)l\text{am}1')$			
Transient - Sheath voltage	V/m	IEEE Std 575	Page 29	Ust	$Ix*X$			
Transient - Circulating current	A	IEEE Std 575	Page 29	Ist	$\text{sqrt}(Ix2(R/Rs)l\text{am}1')$	None	None	None

VOLTAGE DROP

Rated voltage	kV	IEC60840	Table 4	U		33	33	33
Pos Seq Impedance	uohm/m			Z1	$\text{sqrt}(R12+Xm2)$			
Calculated Amps	A	IEC60287-1-1	1.4.1	I	$\text{sqrt}(Etop/Ebot)$	309	294	381
Permissible limit	%	Input		a		2.5%	2.5%	2.5%
Permissible voltage drop	V			E	$a*U$	825	825	825
Voltage drop factor	mV/A/m			R	$\text{sqrt}3*Z$			
Max permissible length	km			L	$E/(I*R)$			

CRITICAL LENGTH

Critical Length	km			Lmax	$\text{sqrt}3(I/wCU)x103$	None	None	None
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RECEIVED POWER

Max Feeding Power	MVA	IEC60287-1-1	1.4.1	S				
Max Received Power	MW			P	$\text{sqrt}[S2-Q2]$			
Power Losses	%			Ploss	P/S	%	%	%
